SOFE 3290 - Operating Systems

Virtual Memory Manager

Lab 6
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CRN74291 - Group 40

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Translating a Logical Address to a Physical Address

The translation process begins when the system reads a 32-bit logical address from standard input. Only the rightmost 16 bits are relevant as the virtual address space is 65,536 bytes. Through bitwise masking and shifting, these 16 bits are split into an 8-bit page number and an 8-bit page offset.

The system first consults the Translation Lookaside Buffer (TLB), a fast hardware cache simulation. If the page number is found here (a TLB hit), the corresponding frame number is instantly retrieved. If it is not found (a TLB miss), the system falls back to checking the main Page Table. If the page table contains a valid frame number, it is retrieved. If the page table entry is invalid, a page fault is triggered to fetch the data from the hard drive.

Finally, the physical address is calculated using the formula:

$$Physical\ Address\ =\ (Frame\ Number\ \times\ Frame\ Size)\ +\ Offset$$

The system then accesses the physical memory array at this calculated address to retrieve the stored signed byte value.

Handling Page Faults and Demand Paging

A page fault occurs when a logical address references a page that is not currently loaded into physical memory (indicated by a -1 in the page table). To resolve this, the program implements demand paging.

When a fault is triggered, the program accesses the simulated hard drive, `BACKING_STORE.bin` . Using the `fseek()` function, it navigates to the exact byte offset of the missing page. It then uses `fread()` to read a 256-byte chunk of data and loads it into the next available open frame in the physical memory array. Once the data is loaded into RAM, the program updates the Page Table with the new frame number and adds the translation to the TLB so future accesses are resolved quickly.

Core Data Structures

The manager relies on three primary data structures simulated using standard C arrays:

- **Physical Memory:** A 65,536-element array of signed char. It acts as the system RAM, logically divided into 256 frames of 256 bytes each.
- **Page Table:** A 256-element integer array mapping page numbers (the array index) to physical frame numbers (the stored value). It is initialized entirely with -1 to indicate that all pages are initially empty.
- **Translation Lookaside Buffer (TLB):** An array of 16 custom struct elements, where each struct holds a page_number and a frame_number. It utilizes a First-In, First-Out (FIFO) replacement policy; a tlb_head counter tracks the oldest entry and overwrites it when the buffer is full.

Performance Statistics and Analysis

Upon processing the entirety of addresses.txt, the program yielded the following performance metrics:

- **Page Fault Rate:** 0.244 (24.4%)
- **TLB Hit Rate:** 0.054 (5.4%)

The TLB hit rate is notably low. This is the expected behavior for this specific simulation because the logical addresses in the provided input file were generated completely at random. In a real-world operating system, programs tend to access the same memory regions repeatedly, which makes the TLB highly efficient. In this case, the TLB frequently misses as this system is not in place.